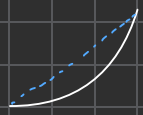
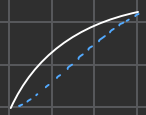


5.1: The shape of curves:

$f'(x) > 0 \Rightarrow f(x)$ is increasing
$f'(x) < 0 \Rightarrow f(x)$ is decreasing

$f''(x) > 0 \Rightarrow$ minimum / concave upwards
$f''(x) < 0 \Rightarrow$ maximum / concave downwards


Concave Upwards


Concave Downwards

Point of inflection $\Rightarrow f''(x) = 0$ (NOT THE CONVERSE!)
To see if there is a point of Inflection, look for a change in concavity / check $f'(x)$ does not change sign.

5.2: The Chain Rule.

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} \quad / \quad f(x) = g(h(x)) \Rightarrow f'(x) = g'(h(x)) \times h'(x)$$

Used to show differentiation in composite functions

Example:

$$y = (2x^4 + 8x)^3 \Rightarrow \frac{dy}{dx} = 3(2x^4 + 8x)^2 \times (8x^3 + 8)$$
$$= (24x^3 + 24)(2x^4 + 8x)^2$$

5.3: Connected Rates of Change

$$\Rightarrow \frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}$$

$A = \pi r^2 \quad \frac{dr}{dt} = 2$

$$\Rightarrow \frac{dA}{dr} = 2\pi r$$
$$\Rightarrow \frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt} = \underline{4\pi r}$$

5.4: Product & Quotient Rules:

$$y = uv \Rightarrow \frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

$$y = \frac{u}{v} \Rightarrow \frac{dy}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$